

Quiz 10 — Imbalanced Data

Compute precision / recall / specificity, choose AUROC vs. AUPRC, match a treatment, and interpolate a SMOTE point

Instructions: Answer all four questions — every part is answerable from the formulas and pictures on the slides. A rapid resistance screen was run on **500 isolates** (positive = *resistant*); Q1–Q2 use its confusion matrix. Compute each rate exactly; no calculators needed.

Name: _____

Date: _____

Formulas (from the slides): recall (sensitivity) = $\frac{TP}{TP + FN}$ specificity = $\frac{TN}{TN + FP}$ precision = $\frac{TP}{TP + FP}$

$$FPR = \frac{FP}{FP + TN}$$

1. Three rates by hand. The screen's confusion matrix (rows = the model's call, columns = truth):

	truth: + (resistant)	truth: - (susceptible)	
call: +	TP = 16	FP = 32	
call: -	FN = 4	TN = 448	
	20 resistant	480 susceptible	500 total

(a) recall (sensitivity) = _____

(b) specificity = _____

(c) precision = _____

(d) A model that ALWAYS calls “susceptible” scores accuracy = $480/500 = 96\%$. The lab must not miss resistant isolates. *Which rate(s) do you trust here, and why is accuracy misleading?*

2. AUROC or AUPRC? Same screen. Positives are **rare: 20 of 500 = 4% prevalence**. Plot this one operating point on each curve, then choose what to report across all thresholds.

(a) ROC point: $FPR = \frac{32}{32 + 448} = \frac{32}{480}$, TPR (recall) = _____ — does it hug the top-left corner? _____

(b) PR point: recall = _____, precision = _____; the PR chance baseline sits at the prevalence = _____

(c) To report this screen across *all thresholds*, would you trust the **AUROC** or the **AUPRC / PR curve**, and why?

3. Match the treatment. Match each situation to *one* move from the menu: **curation** · **class weights** / **SMOTE** / **oversample** · **focal loss** · **curriculum**. Each is used once.

(a) Junk runs and mislabeled spectra contaminate the training set — and the same isolate appears three times. → _____ (b) Only 41 real resistant spectra; the model just predicts “susceptible”. → _____ (c) Easy cases are fine, but borderline isolates are missed. → _____ (d) Training is unstable when very easy and very hard batches alternate. → _____

(e) Which of these four adds NEW resistant signal to the dataset?

4. SMOTE by hand. SMOTE makes a new minority point on the segment between two real ones: $\text{synthetic} = A + \lambda(B - A)$. Use two m/z -bin intensities $A = (2, 6)$ and $B = (4, 10)$.

(a) With $\lambda = 0.25$: $\text{synthetic} = (2 + 0.25(4 - 2), 6 + 0.25(10 - 6)) =$ _____

(b) One sentence: why is raw-spectrum SMOTE risky in 6,000 dimensions?

Keep this sheet — it’s your checklist for any rare-class assay in your own lab.